සියලු ම හිමිකම් ඇවිරිණි / முழுப் பதிப்புரிமையுடையது / All Rights Reserved ]

ශ්‍රී ලංකා විභාග දෙපාර්තමේන්තුව
இலங்கைப் பரீடசைத் திணைக்களம்
Department of Examinations, Sri Lanka

අධ්‍යයන පොදු සහතික පත්‍ර (උසස් පෙළ) විභාගය, 2011 අගෝස්තු
கல்விப் பொதுத் தராதரப் பத்திர (உயர் தர)ப் பரீடசை, 2011 ஓகஸ்ற்
**General Certificate of Education (Adv. Level) Examination, August 2011**

තොරතුරු හා සන්නිවේදන තාක්ෂණය I
தகவல், தொடர்பாடல் தொழினுட்பவியல் I
**Information & Communication Technology I**

**20 E I**

පැය දෙකයි
இரண்டு மணித்தியாலம்
*Two hours*

**Instructions:**

- *Answer **all** the questions.*
- *Write your **Index Number** in the space provided in the answer sheet.*
- *Instructions are also given on the back of the answer sheet. Follow those carefully.*
- *In each of the questions **1** to **50,** pick one of the alternatives from* (1), (2), (3), (4), (5) *which is **correct or most appropriate** and **mark your response on the answer sheet with a cross** (x) **in accordance with the instructions given on the back of the answer sheet.***
- *Use of calculators is **not allowed.***

**1.** Who invented the Analytical Engine?
(1) Blaise Pascal (2) Charles Babbage (3) John Von Neumann
(4) John V. Atanasoff (5) John Presper Eckert

**2.** Which of the following technologies is used in the Second Generation Computers?
(1) Integrated Circuits (ICs) (2) Large Scale Integration (LSI)
(3) Microprocessors (4) Transistors
(5) Vacuum Tubes

**3.** Answer sheet of a multiple choice question paper can be read with
(1) Digitizer. (2) Light pen. (3) MICR. (4) Scanner. (5) POS terminal.

**4.** Computerized medical imaging systems which use echoes of transmitted pulses are called
(1) CAT scans. (2) CT scans. (3) PETT scans.
(4) Ultrasound scans. (5) X-ray images.

**5.** The binary number equivalent to the $27_{10}$ is
(1) 000111 (2) 111000 (3) 011011 (4) 101011 (5) 111010

**6.** The abbreviation GSM stands for
(1) Global System for Mobile Access. (2) Global System for Mobile Communication.
(3) Global System for Mobile Phone. (4) Global System for Mobile Transmission.
(5) Global System for Mobile Interconnection.

**7.** Which of the following database types is widely used in computer applications?
(1) Flat file (2) Hierarchical (3) Network (4) Relational (5) Object oriented

**8.** The main role of a "router" in computer network is
(1) to broadcast data packets to all hosts in the network.
(2) to monitor the network for malicious activities.
(3) to retransmit the received signal at a higher power.
(4) to allow hosts in two networks to communicate.
(5) to allow load balancing between multiple separate networks.

**9.** A command that can be used to check network connectivity to a computer is
(1) ipconfig (2) ping (3) traceroute (4) netstat (5) hostname

**10.** Which of the following best describes the order of the testing conducted in the software development process?
(1) Acceptance testing, integrated testing, unit testing.
(2) Integrated testing, acceptance testing, unit testing.
(3) Integrated testing, unit testing, acceptance testing.
(4) Unit testing, acceptance testing, integrated testing.
(5) Unit testing, integrated testing, acceptance testing.

[See page two

**11.** Which of the following is a classification of systems?
(1) Close and hybrid (2) Natural and structured (3) Open-and close
(4) Open and structured (5) Structured and hybrid

**12.** A banking system could be best considered as a/an
(1) embedded system. (2) enterprise resource planning system.
(3) expert system. (4) knowledge management system.
(5) transaction processing system.

**13.** Which of the following best describes the home page of a website?
(1) The index of the website (2) The welcome page
(3) A description of the website (4) The first page of the website
(5) A web page that presents short journal entries

**14.** A manager of an engineering manufacturing company is considering whether to recruit more people in the near future. What is the most relevant information in this case he has to consider?
(1) Personnel records of all the employees
(2) Previous reports of all the recruited people
(3) The attendance sheets of all the employees
(4) Detailed reports of engineering specifications and manufacturing orders
(5) Summary reports of present and projected personnel requirements

**15.** $C1A_{16} + 4A2_{16} =$
(1) $523_{16}$ (2) $FBC_{16}$ (3) $FBB_{16}$ (4) $0BC_{16}$ (5) $10BC_{16}$

**16.** What is the main function of a DHCP server?
(1) Allocating IP addresses.
(2) Resolving domain names from IP addresses.
(3) Providing directory services to users.
(4) Sharing an Internet connection among users.
(5) Protecting a computer network from virus attacks.

**17.** Consider the IP address 192.248.87.3 and the subnet mask 255.255.255.224. How many hosts can be directly connected to this network?
(1) 16 (2) 24 (3) 30 (4) 64 (5) 128

**18.** What is the most suitable media to transmit a high definition video over a distance of 5 km?
(1) Radio waves (2) Twisted pair copper cables (3) Fibre optics cables
(4) Coaxial cables (5) Open wire cables

**19.** The transport layer of the OSI reference model provides
(1) error correction. (2) routing of data packets.
(3) flow control. (4) process-to-process communication.
(5) error detection.

**20.** Which of the following Python program segment is syntactically correct?

(1)
```
total = 0
for i in range(1, 12):
total = total + i
```

(2)
```
total
for i in range(1, 12):
total = total + i
```

(3)
```
total = 0
for i in range(1, 12):
   total = total + i
```

(4)
```
total = 0
for i in range(1, 12)
   total = total + i
```

(5)
```
total = 0
for i in range(1, 12):
   total = total + i
```

[*See page three*

**21.** Consider the given flow chart:

Which of the following statements about the algorithm represented by the flow chart is **incorrect**?

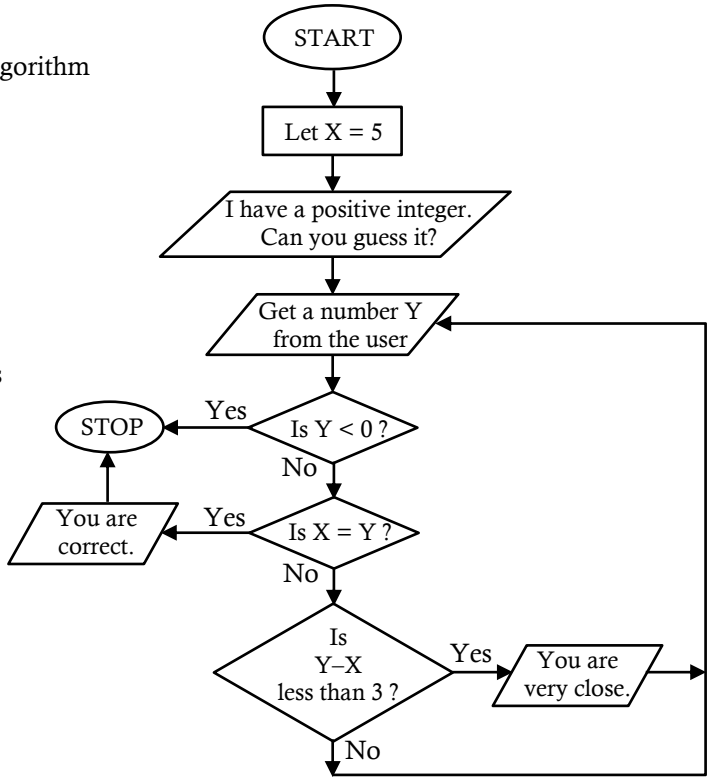
(1) It stops when the user enters the value –1.

(2) It stops when the user enters the value 5.

(3) It gives the message "You are very close." when the user enters the value 9.

(4) It gives the message "You are very close." when the user enters the value 6.

(5) If the value entered by the user is 8, it allows the user to enter another value.




**22.** Which of the following statements best describes the result of hard disk 'fragmentation'?

(1) Hard disk data access speed is reduced.
(2) Network access speed becomes slow.
(3) Hard disk becomes totally inaccessible.
(4) Some data will get erased from the hard disk.
(5) Number of bad sectors get increased.

**23.** Consider the following statements about computerized databases:

A - Need more human resources to manage the computerized database than a manual system.
B - Retrieval of data is efficient than a manual system.
C - No data duplications.
D - Need more space to store data than a manual system.

Which of the above statements are correct with respect to a properly designed database?

(1) A and B only.
(2) A and D only.
(3) B and C only.
(4) B and D only.
(5) C and D only.

- Consider the following system description and the relations A, B, C and D given below to answer the questions **24**, **25** and **26**.

A principal of a National school wants to develop a database to maintain Admission Number, Student Name, Address, National Identity Card number (NIC) and the Date of Birth (DOB) of Advanced Level students. The principal also wants to know the marks obtained for each subject by the students. In addition to the above requirements, the principal needs to know the subjects assigned to the teachers.

Relations :

A - Student(admissionNo, studentName, address, DOB, NIC)
B - Subject(subjectCode, subjectName)
C - Mark(admissionNo, subjectCode, marksObtained)
D - Teacher(teacherNo, subjectCode, teacherName, subjectName, class)

**24.** Which of the above relations are in the third normal form?

(1) A and C only.
(2) A and D only.
(3) A, B and C only.
(4) A, C and D only.
(5) B, C and D only.

**25.** Which of the following combinations of attributes provides the minimal set of primary keys for the relations Student, Subject and Mark respectively?

(1) admissionNo and NIC, subjectCode, admissionNo.
(2) NIC, subjectCode, subjectCode.
(3) admissionNo, subjectCode, subjectCode.
(4) admissionNo, subjectCode, admissionNo and subjectCode.
(5) admissionNo, subjectName, admissionNo and subjectCode.

[*See page four*

**26.** Which of the following SQL statements would produce an output with admission number, name of the student, subject code and marks obtained?

(1) SELECT studentName, subjectCode, marksObtained
FROM Student, Mark
WHERE Student.admissionNo = Mark.admissionNo

(2) SELECT Student.admissionNo, studentName, subjectCode, Student.marksObtained
FROM Student, Mark
WHERE Student.admissionNo = Mark.admissionNo

(3) SELECT Student.admissionNo, studentName, subjectCode, marksObtained
FROM Student, Mark
WHERE Student.admissionNo = Mark.admissionNo

(4) SELECT Student.admissionNo, studentName, subjectCode, marksObtained
FROM Student, Mark
WHERE Student.admissionNo = admissionNo

(5) SELECT Student.admissionNo, studentName, subjectCode, marksObtained
FROM Student, Mark
WHERE admissionNo = Mark.admissionNo

**27.** Which of the following would be the result if the Boolean expression $F(x,y) = (\overline{x+y}).(\overline{\bar{x}+\bar{y}})$ is simplified by using De Morgan's Law?

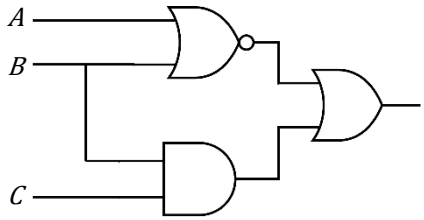
(1) $x$ (2) $y$ (3) 0 (4) 1 (5) $x, y$

**28.** Which of the following Boolean expressions represents the given logic circuit?

(1) $(\overline{A+B}) + (B+C)$ (2) $(A+B) + (B.C)$

(3) $(\overline{A+B}) + (\overline{B+C})$ (4) $(\overline{A.B}) + (B.C)$

(5) $(\overline{A+B}) + (B.C)$




**29.** Consider the following properties :

A - Density
B - Capacity
C - Security
D - Cost
E - Access time

Which of the above are being used as the main properties to classify different types of memory?

(1) B and D only. (2) A, B and C only. (3) B, D and E only.
(4) A, B, C and D only. (5) A, B, D and E only.

**30.** Consider the following memory types:

A - Read Only Memory
B - Secondary storage
C - Cache memory
D - Flash memory
E - Random Access Memory

Which of the above can be considered as Volatile Memories?

(1) A and B only. (2) A and C only. (3) C and D only.
(4) C and E only. (5) D and E only.

**31.** Consider the following statements:

A - Customer can obtain the services while he/she is at his/her home or office.
B - Payments can be made without using currency notes.
C - Customer is open to a large international service provider base.
D - Service provider is open to a larger international customer base.
E - Security of payments are always guaranteed.

Which of the above are advantages in e-commerce compared to traditional commerce?

(1) A, B and C only. (2) A, B and D only. (3) A, C and D only.
(4) A, B, C and D only. (5) B, C, D and E only.

[*See page five*

**32.** Consider the following statements with respect to on-line services:

A - Customer should have access to on-line services.
B - Customer should possess an electronic mechanism to make payments.
C - At the time of placing the order for an item, the identify of the customer should be established.
D - Service provider should provide on-line services.
E - Both the customer and the service provider should be in the same geographical region.

Which of the above items are essential in on-line ordering?

(1) A and B only. (2) A, B and C only. (3) A, B and D only.
(4) A, C, D and E only. (5) B, D and E only.

**33.** Consider the following statements:

A - Reading time from memory (latency) was assumed to be negligible.
B - Utilization of a hierarchical memory structure.
C - Inability to foresee the limitation of the processor clock frequency.

Which of the above prevented the technological advancement beyond the Von Neumann architecture?

(1) A only. (2) B only. (3) C only.
(4) A and B only. (5) A and C only.

**34.** Consider the following computer applications:

A - Guiding Tourists based on their current location.
B - Customization of automobiles.
C - Advertising via Internet.
D - Customization of domestic environments.

Which of the above applications are most favourably benefited from ubiquitous computing?

(1) A and B only. (2) B and C only. (3) A, B and C only.
(4) A, B and D only. (5) B, C and D only.

**35.** Consider the following items:

A - Object oriented
B - Rapid Application Development
C - Spiral
D - Structured
E - Waterfall

"....., ..... and ..... are system development methodologies".

Which of the above are most appropriate respectively to fill the blanks in the above statement?

(1) A, B and C only. (2) A, C and D only. (3) B, C and D only.
(4) B, C and E only. (5) B, D and E only.

**36.** Consider the following statements about the Internet:

A - The Internet is a global network of networks.
B - People and organizations who are connected to the Internet can access its massive store of shared information.
C - W3C is in charge of the Internet.
D - Data can be downloaded only with File Transfer Protocol (FTP).
E - Anybody can publish information or create new services on the Internet.

Which of the above statements are correct?

(1) A, B and D only. (2) A, B and E only. (3) A, D and E only.
(4) B, C and D only. (5) B, C and E only.

**37.** Consider the following Python data items:

A - 15.2
B - [12, 'abc', 5.2]
C - {'name' : 'Nimal', 'age' : 18}

Python data types of the above data items A, B, C are respectively

(1) float, list, dictionary. (2) integer, tuple, dictionary. (3) float, list, tuple.
(4) integer, tuple, list. (5) float, tuple, dictionary.

[*See page six*

**38.** Select the correct layout corresponding to the following code segment of an HTML document.

```
<ul>
<li>Fruits
<ul><li>Mango
<ul>
<li>Gira amba</li>
<li>Dampara</li></ul></li>
<li>Pineapple</li></li></ul>
<li>Vegetables</li>
</ul>
```

(1)
- Fruits
- Mango
  - Gira amba
  - Dampara
  - Pineapple
- Vegetables

(2)
- Fruits
  - Mango
    - Gira amba
    - Dampara
  - Pineapple
- Vegetables

(3)
- Fruits
  - Mango
  - Gira amba
  - Dampara
  - Pineapple
- Vegetables

(4)
- Fruits
- Mango
- Gira amba
- Dampara
- Pineapple
- Vegetables

(5)
- Fruits
  - Mango
  - Gira amba
  - Dampara
  - Pineapple
- Vegetables

**39.** Consider the following items:

A - radio
B - textarea
C - select
D - checkbox
E - textbox

Which of the above can go as attributes of an input element in a form?

(1) A, B and C only. (2) A, B and D only. (3) A, D and E only.
(4) B, C and D only. (5) B, D and E only.

**40.** What is the correct syntax to be used to insert a JavaScript into an HTML page?

(1) `<javascript>` (2) `<javascript language="text/javascript">`
(3) `<script type="text/javascript">` (4) `<scripting language="javascript">`
(5) `<scripting type="javascript">`

**41.** Consider the following XML scripts:

A -
```
<?xml version="1.0"?>
<students>
<name>Shashi Dias</name>
<age>18</age>
<regNo>A25849</regNo>
</students>
```

B -
```
<?xml version="1.0"?>
<students>
<name age="18">Shashi Dias</name>
<regNo>A25849</regNo>
</students>
```

C -
```
<?xml version="1.0"?>
<name>Shashi Dias</name>
<age>18</age>
<regNo>A25849</regNo>
```

D -
```
<xml version="1.0"?>
<students>
<name age=18>Shashi Dias</name>
<regNo>A25849</regNo>
</students>
</xml>
```

Which of the above scripts are syntactically correct?

(1) A and B only. (2) A and C only. (3) A and D only.
(4) B and C only. (5) C and D only.

[*See page seven*

**42.** Consider the following statements about operating systems:

A - Ubuntu is an open source operating system.
B - Windows XP is a proprietary operating system.
C - Linux is a proprietary operating system.

Which of the above statements is/are correct?

(1) A only. (2) B only. (3) C only.
(4) A and B only. (5) A and C only.

**43.** Consider the following statements with respect to ER diagrams:

A - An ER diagram has entities and relationships.
B - Cardinality of all relationships should always be one-to-one.
C - Entities may have attributes.
D - There could be binary and tertiary relationships.

Which of the above statements are correct?

(1) A and D only. (2) B and C only. (3) B and D only.
(4) A, C and D only. (5) B, C and D only.

**44.** Consider the following statements about language translators:

A - Interpreters convert the entire source program into object program at once.
B - Compilers convert the entire source program into object program at once.
C - Programs written in high level languages do not need language translators to execute on a typical computer.

Which of the above statements is/are correct?

(1) A only. (2) B only. (3) C only.
(4) A and B only. (5) B and C only.

**45.** Consider the following statements about programming languages:

A - C programming language is a first-generation language (1GL).
B - Assembly language is a second-generation language (2GL).
C - Python is a second-generation language (2GL).

Which of the above statements is/are correct?

(1) A only. (2) B only. (3) C only.
(4) A and B only. (5) B and C only.

**46.** Consider the following statements in a program:

A - # This is a comment.
B - // This is a comment.
C - /* This is a comment. */
D - a=1 # This is a comment.
E - # Initial value of a=1

Which of the above are syntactically correct Python statements?

(1) A and D only. (2) C and E only. (3) A, D and E only.
(4) B, C and D only. (5) C, D and E only.

**47.** Consider the following assignment statements:

A - a = 'Nimal's address'
B - a = "Nimal's address"
C - a, b, c = 1
D - a, b, c = 1, 2, 'string'
E - a = b = 1

Which of the above are syntactically correct Python statements?

(1) A and C only. (2) B and D only. (3) A, C and E only.
(4) B, D and E only. (5) C, D and E only.

[*See page eight*

**48.** Consider the following incomplete Python program:

```
data=[5, 1, 23, 10]
datacount = len(data)
for i in range(datacount-1):
  for k in range (i, datacount):
.....................................
    temp = data[i]
    data[i],data[k] =data[k],temp
for i in range(datacount):
    print (data[i])
```

Which of the following Python statements should be included at the blank line to print the data items in the data structure named 'data' in the ascending order.

(1) `if data[i] < data[k]:` (2) `if data[i] > data[k]:` (3) `if data[i] = data[k]:`
(4) `if data[i] < data[k]` (5) `if data[i] > data[k]`

**49.** Consider the following Python program:

```
data = [5, 1, 23, 10, -3]
def fun(a):
     i, c=1, a[0]
     while i < len(a):
        if(a[i] > c):
              c=a[i]
        i=i+i
     return c
print (fun(data))
```

Which of the following is the output of this program?

(1) –3 (2) 1 (3) 5 (4) 10 (5) 23

**50.** Consider the following Python program:

```
f1 = open('input.txt','r')
f2 = open('output.txt', 'w')
line = f1.readline()
while (line) :
        data = (line.strip()).split(",")
        total = float(data[1]) + float(data[2])
        f2.write('{},{},{},{}\n'.format(data[0],data[1],data[2],total))
        line = f1.readline()
f1.close()
f2.close()
```

The content of the file "input.txt" is given below.

```
Nimal,30,60
Saman,80,45
Upali,100,80
```

After executing the program what would be the content of the file "output.txt"?

(1)
```
Nimal
Saman
Upali
```

(2)
```
Nimal,30,60
Saman,80,45
Upali,100,80
```

(3)
```
Nimal,30,60,90
Saman,80,45,125
Upali,100,80,180
```

(4)
```
Nimal,30,60,90.0
Saman,80,45,125.0
Upali,100,80,180.0
```

(5)
```
Nimal,30,60,90.0 Saman,80,45,125.0 Upali,100,80,180.0
```

* * *

සියලු ම හිමිකම් ඇවිරිණි / முழுப் பதிப்புரிமையுடையது / All Rights Reserved ]

ශ්‍රී ලංකා විභාග දෙපාර්තමේන්තුව
இலங்கைப் பரீடசைத் திணைக்களம்
Department of Examinations, Sri Lanka

අධ්‍යයන පොදු සහතික පත්‍ර (උසස් පෙළ) විභාගය, 2011 අගෝස්තු
கல்விப் பொதுத் தராதரப் பத்திர (உயர் தர)ப் பரீடசை, 2011 ஓகஸ்ற்
**General Certificate of Education (Adv. Level) Examination, August 2011**

තොරතුරු හා සන්නිවේදන තාක්ෂණය II
தகவல், தொடர்பாடல் தொழினுட்பவியல் II
**Information & Communication Technology II**

**20 E II**

පැය තුනයි
மூன்று மணித்தியாலம்
*Three hours*

**Index No. :** .....

**Important:**

- *This question paper consists of* **10** *pages.*
- *This question paper comprises of two parts,* **Part A** *and* **Part B.** *The time allotted for* ***both parts*** *is* ***three hours.***
- *Use of calculators is* ***not allowed***

**PART A – Structured Essay:**
**(pages 02-07)**

- *Answer* ***all*** *the questions* ***on this paper itself.*** *Write your answers in the space provided for each question. Note that the space provided is sufficient for your answers and that extensive answers are not expected.*

**PART B – Essay:**
**(pages 08-10)**

- *This part consists* ***six*** *questions, of which,* ***four*** *are to be answered. Use the papers supplied for this purpose.*
- *At the end of the time allotted for this paper, tie the* ***two parts together*** *so that* **Part A** *is* ***on top*** *of* **Part B** *before handing them over to the Supervisor.*
- *You are permitted to remove* ***only*** **Part B** *of the question paper from the Examination Hall.*

**For Examiner's Use Only**

| For the Second Paper | | |
|---|---|---|
| **Part** | **Question No.** | **Marks** |
| A | 1 | |
| | 2 | |
| | 3 | |
| | 4 | |
| B | 1 | |
| | 2 | |
| | 3 | |
| | 4 | |
| | 5 | |
| | 6 | |
| **Total** | | |

**Final Marks**

| | |
|---|---|
| In numbers | |
| In words | |

**Code Number**

| | |
|---|---|
| Marking Examiner 1 | |
| Marking Examiner 2 | |
| Marks checked by : | |
| Supervised by : | |

[See page two

## Part A – Structured Essay

*Answer **all four** questions on this **paper itself***

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**1.** (a) State the main technologies used in the first four generations of computers.

(b) Draw a diagram to depict the fetch-execute cycle used in program execution.

[*See page three*

Do not write in this column

(c) Show how the computation 5+(-3) is done in 8-bit two's complement arithmetic. Explain how you deal with the carry generated from the most significant bit.

**2.** (a) Encircle the most suitable entry in the second and third columns corresponding to the properties listed in the first column of the following table with respect to FAT32 and NTFS file systems.

| | FAT32 | NTFS |
|---|---|---|
| Maximum file size | limited/unlimited | limited/unlimited |
| Maximum file name length | limited/unlimited | limited/unlimited |
| Security | yes/no | yes/no |
| Support of Unicode | yes/no | yes/no |

(b) A computer has an 18-bit virtual memory address space where six bits are used for a page address.

(i) Calculate the total number of pages defined by the above addressing scheme.

(ii) Consider the following virtual memory address:

010111000000111100

What is the page and displacement (Offset) of this address?

[*See page four*

(c) Draw the operating system process transition diagram from process creation to termination.

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**3.** Consider the following scenario.

Students in a school participate in different sports such as volleyball, track and field athletics, table tennis, etc. The principal wants to maintain a registry **with admission number, student name, home address, class,** and **sports** he/she participates. A student can participate in **more than one** sport. For a particular sport, there can be **more than one** student. Each student can participate pre-defined number of hours in a sport.

(a) Draw an ER diagram for the above scenario.

[*See page five*

Do not write in this column

(b) Classify with reasons whether the cardinality of relationship(s) identified in section (a) is one-to-one, one-to-many, or many-to-many.

| Relationship | Cardinality | Reason |
|---|---|---|
| | | |

(c) "ER diagrams do not allow attributes to be assigned on relationships". State whether this statement is true or false. Explain your answer by using the given scenario.

[*See page six*

Do not write in this column

(d) A database designer suggested the following relation for the above system. State **two** weaknesses of this relation and suggest necessary modifications.

| AdmissionNo | StudentName | HomeAddress | Class | SportName |
|---|---|---|---|---|
| | | | | |
| | | | | |

[*See page seven*

Do not write in this column

**4.** (a) Classify the following software as either "system software" or as "application software".

| Software | Classification |
|---|---|
| Linux | |
| Word Processor | |
| Web Browser | |

(b) Computer storage devices can be categorized into three types based on the medium used to store /retrieve data. State the **three** types of media and give an example for each type.

(c) The transaction file in a company's payroll system includes employee number, hours worked department code, and week number. Assume that the system maintains a Employee master table and a Department master table. Encircle the most appropriate validation check for each of the data elements given in the following table.

| Data element | Validation checks |
|---|---|
| employee Number | Presence in Employee master table / Numerical value |
| hours worked | Presence in Employee master table / Range check |
| department code | Presence in Department master table / Range check |
| week number | Length / Range check |

(d) Describe the terms "Video conferencing" and "Copyright".

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[*See page eight*

සියලු ම හිමිකම් ඇවිරිණි / முழுப் பதிப்புரிமையுடையது / All Rights Reserved ]

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இலங்கைப் பரீடசைத் திணைக்களம்
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අධ්‍යයන පොදු සහතික පත්‍ර (උසස් පෙළ) විභාගය, 2011 අගෝස්තු
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தகவல், தொடர்பாடல் தொழினுட்பவியல் II
**Information & Communication Technology II**

**20 E II**

## Part B

* *Answer any **four** questions only.*

**1.** (a) What are the three (3) main components of a Central Processing Unit (CPU) of a typical computer? List the main functions of these three components.

(b) Briefly explain why storage compaction is needed in memory management.

(c) For a file of size 10400 bits, calculate the wastage in file space due to incomplete filling of the last cluster (Assume that a cluster has a size of 512 bytes.)

(d) A digital circuit takes four binary digits as an input, and produces 1 as its output if the decimal value represented by the four binary digits is a **prime number** (number which can only be divided by itself and 1), and 0 otherwise. Assume that all four binary digits represent positive decimal values (No bit is allocated for the sign).

(i) The following truth table is designed to describe the above circuit, in which A, B, C and D represents the four binary inputs from the most significant bit to the least significant bit and F(A,B,C,D) as the output of the circuit. Copy the following truth table onto your answer sheet as it is and complete the output column.

| A | B | C | D | F(A,B,C,D) |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | |
| 0 | 0 | 0 | 1 | |
| 0 | 0 | 1 | 0 | |
| 0 | 0 | 1 | 1 | |
| 0 | 1 | 0 | 0 | |
| 0 | 1 | 0 | 1 | |
| 0 | 1 | 1 | 0 | |
| 0 | 1 | 1 | 1 | |
| 1 | 0 | 0 | 0 | |
| 1 | 0 | 0 | 1 | |
| 1 | 0 | 1 | 0 | |
| 1 | 0 | 1 | 1 | |
| 1 | 1 | 0 | 0 | |
| 1 | 1 | 0 | 1 | |
| 1 | 1 | 1 | 0 | |
| 1 | 1 | 1 | 1 | |

(ii) Write a Boolean expression to represent the logic function of the above circuit in the sum of products form.

(iii) Design a logic circuit for the Boolean expression you have obtained for the above part (ii).

[See page nine

**2.** (a) Describe the terms "elements" and "attributes" with respect to an HTML document.

(b) Identify each of the following as either an element or an attribute and describe their functionality.

(i) br (ii) href (iii) src (iv) html

(c) Consider the following figure which shows a section of a web page of a tour operating company in Sri Lanka.

Answer the following questions using the above figure.

(i) It is required to format all the paragraphs of the above HTML document in "**arial**" font, **14** font size and in **blue** colour. Write the required CSS code segment for the paragraph.

(ii) Explain the effect of having the following tag in the above HTML document.

```
<a href = "elephants.jpg"><img src = "elephants_tnl.jpg"
ALT = "Tour to Yala" width = "288cm" height = "156cm"
style = "border:none"/></a>
```

(iii) Write HTML code segment to create the collection of three radio buttons labelled as 'Blue Whale', 'Leopard' and 'Elephant' as appeared in the above HTML document.

(iv) The company wants to add a table showing the rates as given below with the caption 'Wild Sri Lanka', to the above HTML Document.

| Days | Price |
|---|---|
| 7 | US$910 |
| 10 | US$1220 |

Write HTML code segment to create the table.

[*See page ten*

**3.** (a) You have been asked to design two physically separated networks, namely A and B, each having exactly 10 computers. The IP addresses of A and B networks are **10.32.5.0** and **10.32.6.0** respectively It is required that the computers in the two networks must be able to communicate with each other.

(i) Suggest a suitable subnet mask for each of these networks.

(ii) Name the device required to connect these two physical networks to communicate with each other.

(iii) Draw a network diagram for the above network and assign suitable IP addresses for the devices in these two networks.

(b) (i) Compare TCP and UDP protocols in terms of reliability.

(ii) Peer-to-peer (P2P) and client-server models are distributed application architectures. State the difference between them.

(iii) List the differences between hubs and switches in a network.

**4.** (a) Identify and describe the phases of the waterfall model in software development.

(b) Describe functional and non functional requirements of a system. Identify **two functional** and **three non functional** requirements for a mobile phone.

(c) Describe the purpose of unit, integrated and acceptance testing. Who are the people responsible for each testing process?

(d) Suppose you are planning to buy a new mobile phone and would like to test its functionality. Describe how Black Box testing can be used in this process.

**5.** (a) Explain the necessity of program translators in computer programming.

(b) Give **two** main features for each of the First-Generation and Second-Generation programming languages.

(c) Give **three** main flow control structures used in a structured programming language. Show how these flow control structures can be represented in a flow chart.

(d) The following Python program is intended to convert user given positive integers to their equivalent binary representations. The program should halt when the user inputs the value 0. The program has both syntactic and logical errors. The line numbers are not part of the program, but they are used to reference the lines.

```
1   x = int (input ("Enter an integer →)
2   while x!=0:
3   bn = ""
4     while x > 1:
5        quotient = int(x/2)
6        remainder == x % 2
7        bn = bn + str(remainder);
8           x = quotient
9        bn = str(x) + bn
10    print ("Binary Number", bn)
11    x = int (input("Enter an integer →)
```

(i) State the lines with syntactic errors and state the error.

(ii) Which lines of the program should be changed and state how they should be changed to obtain the desired results. (You are not allowed to add new lines or to delete existing lines.)

**6.** (a) (i) Using an example for each category explain the three types of business: Business to Business (B2B), Business to Consumer (B2C) and Consumer to Consumer (C2C) in e-commerce.

(ii) Chairman of a company is considering fax, e-mail and web as communication tools for a B2E (Business to Employee) application. Being an ICT student recommend the most appropriate tool with reasons.

(b) (i) In the domain of Agent technology, explain the term 'Agent'.

(ii) Give **two** main characteristics of an Agent.

(iii) Briefly explain an example where Agent technology could be used effectively.

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